

HOUSE No. 495

The Commonwealth of Massachusetts

PRESENTED BY:

David M. Rogers

To the Honorable Senate and House of Representatives of the Commonwealth of Massachusetts in General Court assembled:

The undersigned legislators and/or citizens respectfully petition for the adoption of the accompanying bill:

An Act reducing emissions from artificial intelligence.

PETITION OF:

NAME:	DISTRICT/ADDRESS:	DATE ADDED:
<i>David M. Rogers</i>	<i>24th Middlesex</i>	<i>1/17/2025</i>

HOUSE No. 495

By Representative Rogers of Cambridge, a petition (accompanied by bill, House, No. 495) of David M. Rogers relative to reducing emissions from artificial intelligence. Economic Development and Emerging Technologies.

The Commonwealth of Massachusetts

In the One Hundred and Ninety-Fourth General Court
(2025-2026)

An Act reducing emissions from artificial intelligence.

Be it enacted by the Senate and House of Representatives in General Court assembled, and by the authority of the same, as follows:

1 SECTION 1. The General Laws, as appearing in the 2022 Official Edition, are hereby
2 amended by inserting after Chapter 93L the following new chapter:

3 Chapter 93M. Artificial Intelligence

4 Section 1: Definitions

5 (a) As used in this chapter, the following words shall have the following meanings unless
6 the context clearly requires otherwise:

7 “Artificial intelligence,” any technology, including, but not limited to, machine learning,
8 that uses data to train an algorithm or predictive model that enables a computer system or service
9 to autonomously perform any task, including, but not limited to, visual perception, language
10 processing or speech recognition, that is normally associated with human intelligence,
11 perception, or judgment.

“Generative artificial intelligence”, artificial intelligence, including, but not limited to, a general-purpose artificial intelligence model, that is used to produce or edit text, audio, images, or video media.

“Artificial intelligence model”, a component of an information system that implements artificial intelligence technology and uses computational, statistical, or machine-learning techniques to produce outputs from a given set of inputs.

“Artificial intelligence system”, any data system, software, hardware, application, tool, or utility that operates in whole or in part using artificial intelligence.

“Covered entity”, any entity or any person, other than an individual acting in a non-commercial context, that operates a search engine with an artificial intelligence or generative artificial intelligence service. The term “covered entity” does not include:—

government agencies or service providers to government agencies that exclusively and solely process information provided by government entities;

the entity or person’s average annual gross revenues during the period did not exceed \$10,000,000 in the past 3 calendar years.

“Affirmative consent”, an affirmative act by an individual that clearly communicates the individual’s freely given, specific, and unambiguous authorization for an act or practice after having been informed, in response to a specific request from a covered entity that meets the requirements of this chapter.

“Search engine”, a service that utilizes keywords and phrases inputted by an individual to search the internet and provide an index of web pages, images or videos.

“Reporting Entity, any company, organization, or other entity that—
develops or operates an artificial intelligence system; or
owns or operates, in whole or in part, a source of greenhouse gas emissions from a
generator of electricity or a commercial or industrial site that powers artificial intelligence
systems developing or making for use in Massachusetts.

The term “reporting entity” does not include an entity or person whose average annual
gross revenues during the period did not exceed \$10,000,000 once in the past 3 calendar years.

Section 2: Affirmative Consent for Artificial Intelligence

Not later than 18 months after the effective date of this law, no covered entity shall
operate a search engine that automatically returns results using artificial intelligence, provided,
that each individual user may expressly authorize, through affirmative consent, that the covered
entity may utilize artificial intelligence or generative artificial intelligence when returning results
for said user searches

The Executive Office of Technology Services and Security shall recognize one or more
acceptable centralized mechanisms for individuals to exercise affirmative consent for generative
artificial intelligence models in search engines.

(1) Any such affirmative consent mechanism shall:—

(i) require covered entities or service providers acting on behalf of covered entities to
inform individuals about the affirmative consent choice;

- (ii) present the individual with the affirmative consent choice before a generative artificial intelligence result is produced for a search query;
- (iii) provide the individual with a continuous affirmative consent choice for each search query;
- (iv) be consumer-friendly, clearly described, and easy-to-use by a reasonable individual;
- (v) be provided in any covered language in which the covered entity provides products or services subject to affirmative consent; and;
- (vi) be provided in a manner that is reasonably accessible to and usable by individuals with disabilities.

Section 3: Study on the Impacts of Artificial Intelligence

(a) Not later than 18 months after the date of enactment of this Act, the Secretary of the Executive Office of Energy and Environmental Affairs in collaboration with the Secretary of the Executive Office of Technology Services and Security, shall carry out, and submit a report to the Speaker of the House, Senate President, Chair of the Joint Committee on Environment and Natural Resources, Chair of the Joint Committee on Telecommunications Utilities and Energy, and make publicly available the results of, a comprehensive study on the environmental impacts of artificial intelligence.

(1) The study required under this section shall include an examination of:—

- (i) the energy consumption and pollution associated with the full lifecycle of artificial intelligence models, including the design, development, deployment, and use of those artificial intelligence models;

(ii) the energy consumption and pollution associated with the full lifecycle of artificial intelligence hardware, including the extraction of raw materials, manufacturing, and electronic waste associated with that hardware;

(iii) the energy and water consumption for the cooling of the data centers used in the design, development, deployment, and use of artificial intelligence models;

(iv) how choices made during the design, development, deployment, and use of artificial intelligence models, including the efficiency of the artificial intelligence models used, the location, power source, and design of data centers used, and the type of hardware used, impact the resulting environmental effects;

(v) potential environmental impacts that could be acute at local scales, which may include added power loads that create grid stress, water withdrawals that create water stress, or local noise effects;

(vi) the positive environmental impacts associated with applications of artificial intelligence, which may include optimizing systems for energy efficiency, developing renewable energy, enabling discovery of new materials, and automatically monitoring environmental changes;

(vii) the negative environmental impacts associated with applications of artificial intelligence, which may include rebound effects, behavioral impacts, and accelerating high-pollution activities;

(viii) disparate impacts in the negative environmental impacts of artificial intelligence;

(ix) other environmental impacts, as determined by the Secretary of the Executive Office of Energy and Environmental Affairs.

(b) The Secretary of Executive Office of Energy and Environmental Affairs, in consultation with, the Secretary of Energy the Executive Office of Technology Services and Security, and such others as the Secretary's consider appropriate, develop a system for reporting entities of the full range of environmental impacts of artificial intelligence.

(1) Each reporting entity must monitor and track the annual environmental impacts, including the carbon emissions, water usage, production of electronic and other waste, mining of materials, and air, water, and soil pollution, caused by each product and report these annually to the Department of Energy and Environmental Affairs in a format determined by their office.

(2) Each reporting entity must additionally report any efforts to offset or mitigate the impacts of their products, or lack thereof.

(3) These statements shall be made available for public view on the Executive Office of Energy and Environmental Affairs website.

(4) A reporting entity that fails to report its annual environmental impacts shall be subject to a penalty not to exceed \$20,000 for each year it fails to submit the report.